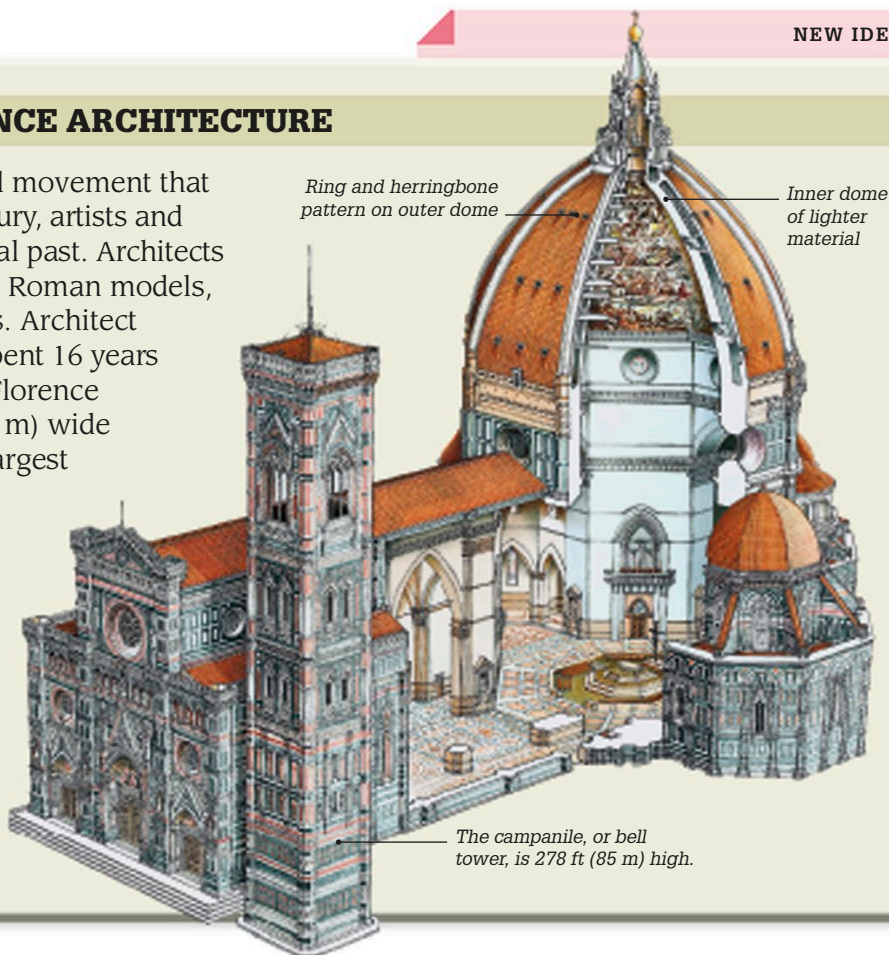


c 1400–1600 RENAISSANCE ARCHITECTURE

During the Renaissance, a cultural movement that began in Italy in the mid-14th century, artists and architects rediscovered the classical past. Architects based their buildings on Greek and Roman models, using columns, arches, and domes. Architect Filippo Brunelleschi of Florence spent 16 years building the remarkable dome of Florence Cathedral. This dome, at 147 ft (45 m) wide and 374 ft (114 m) high, was the largest unsupported dome yet built.

The dome of Florence Cathedral

Building such a huge dome was believed to be impossible. Brunelleschi designed an inner dome of lightweight material and an outer one of heavier stone. Oak timbers set in rings connected the two domes and supported them. Constructing the outer dome was made easier because the builder could balance on the already finished inner dome.



1405

1445

1421

First recorded patent

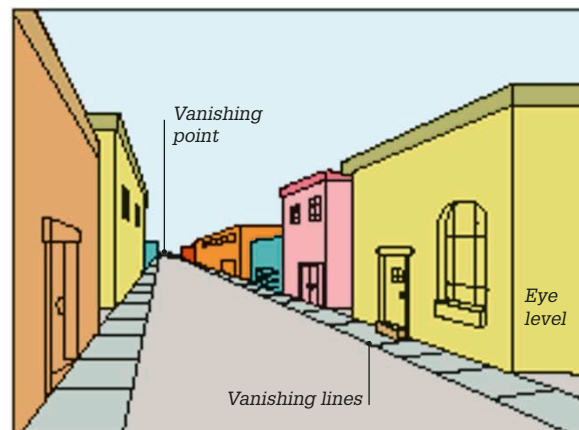
The first patent—a license giving an inventor sole rights to an invention—was granted by the city of Florence to the Italian architect Filippo Brunelleschi. It was for a barge and hoist used to transport heavy marble slabs up the Arno River. The patent forbade anyone else from copying the idea for three years.



1436

Perspective in painting

Roman artists knew how to use perspective (a mathematical system for creating the appearance of distance on a flat surface). Knowledge of the technique was later lost, but rediscovered during the Italian Renaissance. In 1436, Leon Battista Alberti, an architect and scholar, gave a full account of it in his work *On Painting*.



Nicholas of Cusa

German theologian Nicholas of Cusa (1401–1464) believed that all things in the Universe are in motion. From this he concluded that Earth is not fixed and must move around the Sun.

Nicholas of Cusa, portrayed here in a woodcut, had theories on the Universe that would influence scientists in later centuries.

Perspective gives the illusion of depth. The artist draws objects smaller and closer together until eventually they form a single point known as the vanishing point.